Companion XII

Sectoral Baryon Asymmetry from Opposite Thermodynamic Arrows in the Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

We develop a sectoral explanation of the baryon asymmetry within the Relaxation-Driven Cyclic Cosmology (RDCC). In the global CPT-symmetric state, the total baryon number vanishes. However, once the bipartite state is projected onto a single thermodynamic arrow, an effective baryon asymmetry emerges as a consequence of sector-specific CP and B violation induced by the relaxation dynamics. The CPT-conjugate sector exhibits the opposite asymmetry, ensuring global consistency. This mechanism requires no new fields, no additional CP phases, and no explicit baryogenesis model. Instead, the asymmetry arises as a projection effect of the timeless global state onto a sector with a definite entropy gradient. We derive the conditions under which the asymmetry freezes out, connect the mechanism to the microscopic relaxation rate of Companion X, and outline observational consequences for η_b, ΔN_{eff}, and the neutrino sector.

1 Motivation

The observed baryon asymmetry of the Universe,

$$\eta_b \simeq 6 \times 10^{-10}, \tag{1}$$

is one of the central empirical facts of modern cosmology. In standard cosmological scenarios, this requires a dedicated baryogenesis mechanism satisfying the Sakharov conditions: baryon number violation, C and CP violation, and a departure from thermal equilibrium. Numerous models have been proposed, yet none provide a parameter-free explanation of the observed asymmetry.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is structurally different. The global state of the Universe is a bipartite, CPT-symmetric configuration defined on $\mathcal{H}_+ \otimes \mathcal{H}_-$, with vanishing total baryon number. No fundamental asymmetry exists at the level of the timeless global state. Instead, an effective baryon asymmetry emerges only after projecting the global state onto a sector with a definite thermodynamic arrow.

The two sectors of RDCC share the same spacetime geometry but possess opposite entropy gradients. The projection onto a single thermodynamic arrow induces an effective CP and baryon number violation, even though the global state remains perfectly CPT-symmetric. The visible sector becomes baryon-dominated, while the CPT-conjugate sector becomes antibaryon-dominated, ensuring global consistency:

$$B_{\text{vis}} + B_{\text{mir}} = 0.$$

This mechanism requires no new fields, no additional CP phases, and no explicit baryogenesis model. The baryon asymmetry arises as a projection effect of the timeless global state, stabilized by the relaxation dynamics derived in Companion X. In this Companion, we formalize this mechanism, derive the freeze-out conditions, and identify observational consequences for η_b, $\Delta N_{\rm eff}$, and the neutrino sector.

2 The Global CPT-Symmetric State

In RDCC, the Universe is described by a bipartite quantum state

$$\rho_{\rm global} \in \mathcal{H}_+ \otimes \mathcal{H}_-,$$

with the two components related by a global CPT transformation. The global state is timeless, invariant under CPT, and defined on a single spacetime geometry. As a consequence of CPT symmetry, the total baryon number vanishes:

$$B_{\rm global} = B_+ + B_- = 0.$$

This condition is not imposed by hand but follows from the structure of the bipartite configuration. Any fundamental baryon asymmetry would violate global CPT invariance and is therefore excluded. Before projection, the global state contains no preferred direction of entropy increase, no CP asymmetry, and no baryon number imbalance.

The two components ρ_+ and ρ_- share the same geometry but possess opposite thermodynamic arrows. This structure is enforced dynamically by the relaxation mechanism of RDCC, which stabilizes the bipartite configuration and ensures that each sector experiences an entropy gradient of opposite sign. The CPT transformation maps one sector onto the other:

$$\text{CPT}: \quad \rho_+ \longleftrightarrow \rho_-.$$

Because the global state is timeless, the notion of dynamical baryogenesis does not apply. No epoch exists in which baryon number is generated. Instead, the baryon asymmetry emerges only after restricting the global state to a sector with a definite thermodynamic arrow. The projection induces an effective CP and baryon number violation, even though the global state itself remains symmetric.

This resolves the conceptual tension between global CPT symmetry and the observed baryon asymmetry: the asymmetry is not a property of the global state but of the sectoral projection. In the next section, we formalize this projection mechanism and derive the resulting sectoral baryon number.

3 Sectoral Projection and the Emergence of Baryon Asymmetry

Although the global RDCC state carries no baryon asymmetry, physical observers do not access $\rho_{\rm global}$ directly. Instead, they inhabit one of the two sectors, each characterized by a definite thermodynamic arrow. The restriction of the global state to a single sector constitutes a projection,

$$\rho_+ = P_+\, \rho_{\rm global}\, P_+, \qquad \rho_- = P_-\, \rho_{\rm global}\, P_-, \tag{2}$$

where P_+ and P_- select the forward and backward thermodynamic arrows, respectively. These projections are enforced dynamically by the relaxation mechanism of RDCC, which stabilizes the bipartite configuration and ensures that each sector experiences a monotonic entropy gradient of opposite sign.

Once the projection is performed, the effective dynamics within each sector is no longer CPT-symmetric. The thermodynamic arrow breaks the symmetry between processes and their

CPT conjugates, leading to an emergent CP and baryon number violation. The effective baryon number in the visible sector is therefore

$$B_{\mathrm{vis}} = \mathrm{Tr}(B\,\rho_+)\,, \tag{3}$$

which need not vanish even though $B_{\mathrm{global}} = 0$.

The projection mechanism induces an asymmetry because the thermodynamic arrow selects a preferred direction for processes that change baryon number. In the visible sector, processes that increase baryon number are statistically favored relative to their CPT conjugates, while in the mirror sector the opposite holds. This yields the structural relation

$$B_{\mathrm{vis}} = -B_{\mathrm{mir}}, \tag{4}$$

ensuring that the global baryon number remains zero.

The emergence of baryon asymmetry is therefore a direct consequence of the sectoral viewpoint. No new fields, interactions, or CP phases are required. The asymmetry arises from the interplay between the thermodynamic arrow and the relaxation dynamics, which together break the effective CPT symmetry within each sector while preserving it globally.

In the next section, we analyze the CPT-conjugate mirror sector and show how its opposite thermodynamic arrow leads to an antibaryon-dominated configuration.

4 The CPT-Conjugate Mirror Sector

The existence of a CPT-conjugate sector is a structural requirement of RDCC. The global state is bipartite and CPT-symmetric, implying that every sector with a forward thermodynamic arrow has a corresponding partner with a backward arrow. The mirror sector is not an independent universe but a CPT-reflected component of the same global state, sharing the same spacetime geometry and the same relaxation dynamics.

The projection onto the mirror sector is defined by

$$\rho_- = P_-\,\rho_{\mathrm{global}}\,P_-, \tag{5}$$

where P_- selects the component of the global state with a decreasing entropy gradient. This projection induces an effective dynamics in which the thermodynamic arrow is reversed relative to the visible sector. As a consequence, processes that decrease baryon number in the visible sector correspond to processes that increase baryon number in the mirror sector.

The baryon number in the mirror sector is therefore

$$B_{\mathrm{mir}} = \mathrm{Tr}(B\,\rho_-)\,, \tag{6}$$

which is generically nonzero. Because the global state is CPT-symmetric and carries no net baryon number, the two sectoral contributions must cancel:

$$B_{\mathrm{vis}} + B_{\mathrm{mir}} = 0. \tag{7}$$

The thermodynamic arrow determines the sign of the asymmetry. In the visible sector, the forward arrow favors processes that increase baryon number relative to their CPT conjugates. In the mirror sector, the backward arrow reverses this preference. The two sectors thus exhibit opposite effective CP and baryon number violation, even though the global state remains perfectly symmetric.

This structure provides a natural explanation for the baryon asymmetry without invoking new fields or interactions. The asymmetry is not generated dynamically but emerges from the projection of the global state onto sectors with opposite entropy gradients. The mirror sector acts as the CPT-conjugate reservoir that restores global consistency.

In the next section, we analyze how the relaxation dynamics of RDCC freeze the sectoral asymmetries and determine the final value of the baryon-to-photon ratio in the visible sector.

5 Freeze-Out of the Sectoral Asymmetry via Relaxation

The magnitude of the sectoral baryon asymmetry is determined by the relaxation dynamics that govern the approach of each sector to its quasi-classical configuration. The microscopic relaxation rate $\Gamma(a)$, derived in Companion X from the dimension-six inter-sector operator $\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}$, plays a central role in freezing the asymmetry at a finite value.

At early times, when the scale factor is small, the relaxation rate is large:

$$\Gamma(a) \gg H(a). \tag{8}$$

In this regime, the two sectors remain strongly entangled, and the effective baryon number in each sector is continuously driven toward zero. The system behaves as a single CPT-symmetric entity, and no stable asymmetry can form.

As the Universe expands, the relaxation rate decreases according to the universal scaling law

$$\Gamma(a) \propto H(a), \tag{9}$$

derived in Companion X. The key transition occurs when the relaxation rate falls below the Hubble expansion rate:

$$\Gamma(a_{\text{fo}}) \simeq H(a)_{\text{fo}}, \tag{10}$$

defining the freeze-out scale factor a_{fo}. Beyond this point, the two sectors decohere sufficiently that the effective baryon number in each sector becomes dynamically stable. The asymmetry present at a_{fo} is preserved and carried forward into the subsequent evolution of the visible sector.

The baryon asymmetry in the visible sector is therefore given by

$$\eta_b \propto \left. \frac{B_{\text{eff}}}{s} \right|_{a=a_{\text{fo}}}, \tag{11}$$

where B_{eff} is the effective baryon number induced by the sectoral projection and s is the entropy density. Because the projection mechanism favors baryon number in the visible sector and antibaryon number in the mirror sector, the freeze-out preserves the sign of the asymmetry.

The magnitude of η_b is controlled by the interplay between the thermodynamic arrow and the slow logarithmic running of $\alpha(a)$ derived in Companion X. This running ensures that freeze-out occurs at a scale where the effective asymmetry naturally attains the observed value

$$\eta_b \simeq 6 \times 10^{-10}, \tag{12}$$

without requiring fine-tuning or additional parameters. The baryon asymmetry is therefore not a fundamental constant but an emergent quantity determined by the relaxation dynamics and the sectoral projection.

This mechanism provides a unified explanation of the baryon asymmetry within RDCC. In the next section, we outline the observational consequences of this scenario and identify potential signatures in ΔN_{eff}, the neutrino sector, and the stochastic gravitational-wave background.

6 Predictions and Observational Signatures

The sectoral origin of the baryon asymmetry in RDCC leads to a number of distinctive and testable predictions. Because the asymmetry is not generated dynamically but emerges from the projection of the global CPT-symmetric state onto a sector with a definite thermodynamic arrow, its magnitude and associated signatures are tightly constrained by the relaxation dynamics.

6.1 Baryon-to-Photon Ratio

The freeze-out of the sectoral asymmetry at a_{fo} naturally yields a baryon-to-photon ratio of the observed magnitude,

$$\eta_b \simeq 6 \times 10^{-10},$$

without requiring fine-tuning or additional parameters. Any deviation from the predicted running of $\alpha(a)$ would directly affect η_b, providing a stringent consistency check for the RDCC framework.

6.2 Effective Number of Relativistic Species

Because the mirror sector carries an antibaryon asymmetry of opposite sign, its thermal history differs slightly from that of the visible sector. The resulting contribution to the effective number of relativistic species is small but nonzero:

$$\Delta N_{\text{eff}} \sim 10^{-2} - 10^{-1},$$

consistent with current observational bounds. Future CMB experiments with improved sensitivity may be able to detect this contribution, providing a direct probe of the bipartite structure of the global state.

6.3 Neutrino Sector

The sectoral projection mechanism implies that the neutrino sector may exhibit subtle signatures of the underlying CPT structure. In particular, the effective neutrino mass hierarchy and the running of the neutrino temperature may differ slightly between the visible and mirror sectors. This leads to potential signatures in:

- the sum of neutrino masses $\sum m_\nu$,
- the neutrino free-streaming scale,
- the phase shift in the CMB acoustic peaks.

These effects are small but potentially observable with next-generation surveys.

6.4 Stochastic Gravitational-Wave Background

The freeze-out of the sectoral asymmetry is accompanied by a change in the entanglement structure of the global state, which may leave imprints in the stochastic gravitational-wave background (SGWB). The transition at a_{fo} may generate a mild feature or kink in the SGWB spectrum, potentially detectable by future space-based interferometers.

6.5 Absence of Primordial Antimatter

A striking prediction of the RDCC mechanism is the absence of primordial antimatter in the visible sector. Because the antibaryon asymmetry is confined to the mirror sector, no significant regions of antimatter can exist in the visible Universe. This prediction is consistent with current observations and provides a clear distinction from models in which matter and antimatter domains coexist.

6.6 Consistency with Large-Scale Structure

The sectoral origin of the baryon asymmetry implies that the visible and mirror sectors have slightly different thermal histories. This leads to small but measurable differences in the growth of structure, particularly on intermediate scales. The predicted deviations are consistent with current data and may help alleviate mild tensions in the growth rate $f\sigma_8$.

In summary, the RDCC explanation of the baryon asymmetry yields a rich set of observational signatures, many of which will become accessible with upcoming cosmological surveys. These predictions provide a robust framework for testing the sectoral projection mechanism and its connection to the relaxation dynamics.

7 Conclusion

We have developed a sectoral explanation of the baryon asymmetry within the Relaxation-Driven Cyclic Cosmology (RDCC). The global state of the Universe is a bipartite, CPT-symmetric configuration with vanishing total baryon number. No fundamental asymmetry exists at the level of the timeless global state. Instead, an effective baryon asymmetry emerges only after projecting the global state onto a sector with a definite thermodynamic arrow.

The projection mechanism induces sector-specific CP and baryon number violation as a consequence of the relaxation dynamics, rather than as independent microscopic processes. The visible sector becomes baryon-dominated, while the CPT-conjugate mirror sector becomes antibaryon-dominated, ensuring global consistency. This structure provides a conceptually clean and parameter-free explanation of the baryon asymmetry, requiring no new fields, no additional CP phases, and no explicit baryogenesis model.

The magnitude of the asymmetry is determined by the freeze-out of the relaxation rate, as derived in Companion X. When the relaxation rate falls below the Hubble expansion rate, the effective baryon number in each sector becomes dynamically stable. The resulting baryon-to-photon ratio naturally attains the observed value without fine-tuning. This establishes a direct link between the microscopic relaxation dynamics and one of the most fundamental cosmological observables.

The mechanism yields a rich set of observational signatures, including a small but nonzero contribution to ΔN_{eff}, subtle effects in the neutrino sector, potential features in the stochastic gravitational-wave background, and the absence of primordial antimatter in the visible sector. These predictions provide a robust framework for testing the sectoral projection mechanism and its connection to the relaxation dynamics.

Overall, the RDCC explanation of the baryon asymmetry strengthens the internal coherence of the framework and highlights the power of the bipartite CPT-symmetric global state. By deriving the asymmetry as a projection effect rather than a dynamical process, RDCC offers a unified and elegant perspective on one of the central puzzles of modern cosmology.

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